Biology GCE O Level 2025 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0510 Biology 1, June 2025, Ordinary Level, General Certificate of Education Board). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; select only one answer. Non-programmable calculators are allowed.

Multiple choice questions - choose the one best answer, A, B, C or D. Section I (Questions 1-7) and Section II (Questions 8-20, biological diagrams) are independent questions; Section III (21-25) uses the 1, 2, 3 combinations; Section IV (26-30) asks about two statements; Section V (31-40) matches a heading to a description; Section VI (41-50) deals with biological situations.

1. In which organ of the human body are bile salts produced?

a) Pancreas.
b) Duodenum.
c) Gall bladder.
d) Liver.

2. Which of these organisms will occupy the first trophic level in an ecosystem?

a) Cabbage.
b) Caterpillar.
c) Earthworm.
d) Beetle.

3. Which of the following is the causative agent for malaria?

a) Female Anopheles mosquito.
b) Culex mosquito.
c) Plasmodium.
d) Male anopheles mosquito.

4. The streamline body of the bird enables it to:

a) Capture prey easily.
b) Reduce air resistance during flight.
c) Reduce weight during flight.
d) Perch on tree branches with ease.

5. Comparing inspired and expired air, expired air:

a) Is warmer.
b) Contains no oxygen.

c) Is dry.
d) Is richer in nitrogen.

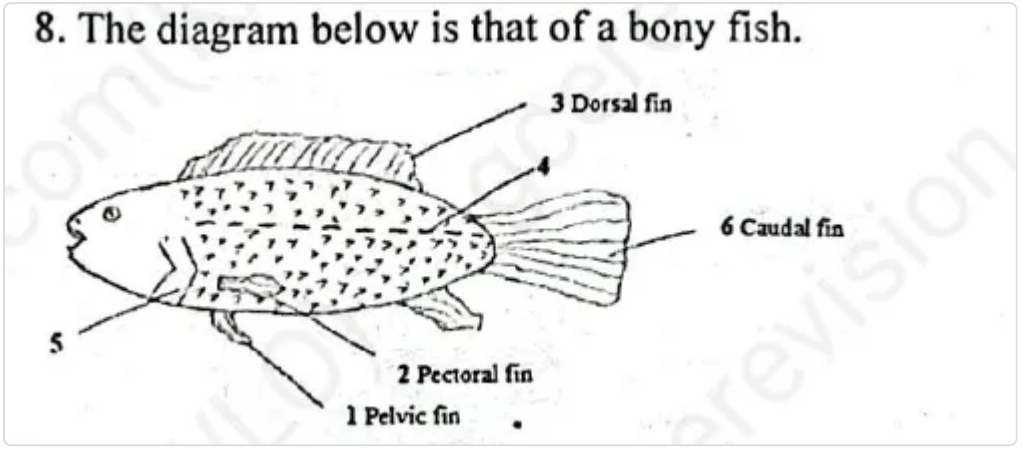
6. Which of the following is an air pollutant?

Options as printed: A) Herbicides. B) Refuse. C) Sewage. D) Noise.

7. A food sample was tested with Millon's reagent and a pink coloration was observed. The food sample contains:

a) Starch.
b) Lipid.
c) Reducing sugar.
d) Protein.

8. The diagram below is that of a bony fish.



Which numbered parts control pitching of fish in water?

a) 1 and 2.
b) 2 and 6.
c) 2 and 3.
d) 6 and 3.

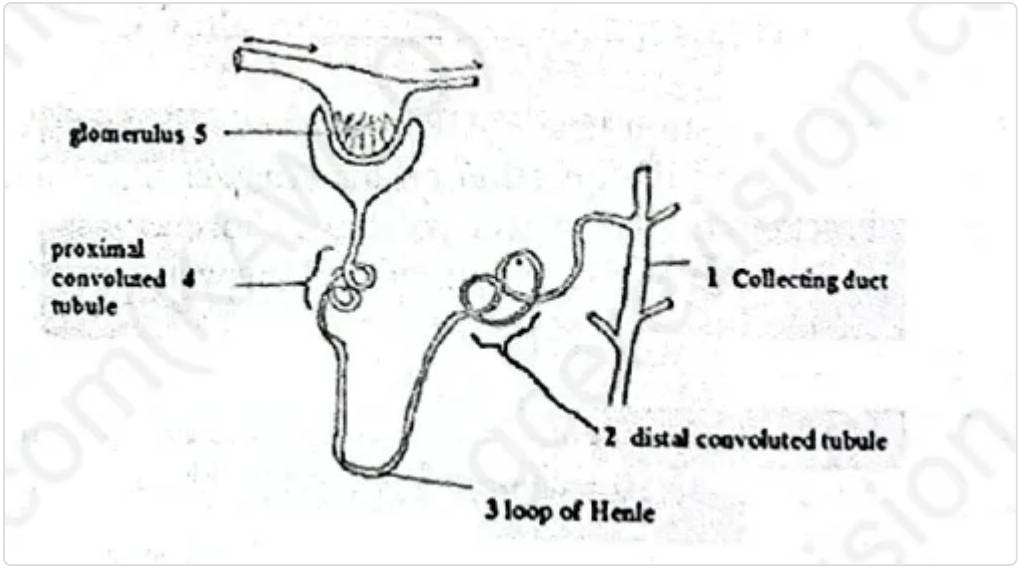
9. The structure in the mammalian eye that contains light sensitive cells is:

a) Lens.
b) Blind spot.
c) Choroid.
d) Retina.

10. How does the structure of the artery help in its function?

a) Thin elastic walls.
b) Thin muscular walls.
c) Thick elastic muscular walls.
d) Thick inelastic muscular walls.

11. The diagram below represents a nephron



Where does ultra-filtration occur?

a) 4.
b) 5.
c) 2.
d) 1.

12. Where does gaseous exchange take place in the human respiratory system?

a) Bronchioles.
b) Alveoli.
c) Trachea.
d) Bronchus.

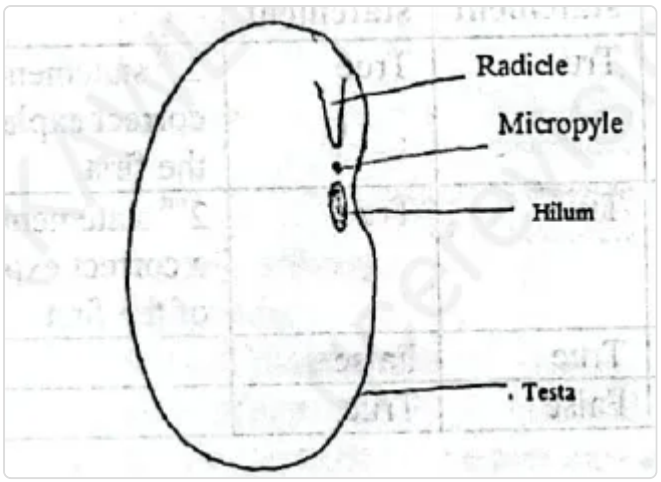
13. Why is it that only 10% energy is transferred from one trophic level to another?

a) Energy is stored in the organisms.
b) Energy is released by the organisms.
c) Energy is burnt up.
d) Energy is used in metabolic processes.

14. Which structure in a bird plays a role in food storage?

a) Crop.
b) Gizzard.
c) Bill.
d) Intestines.

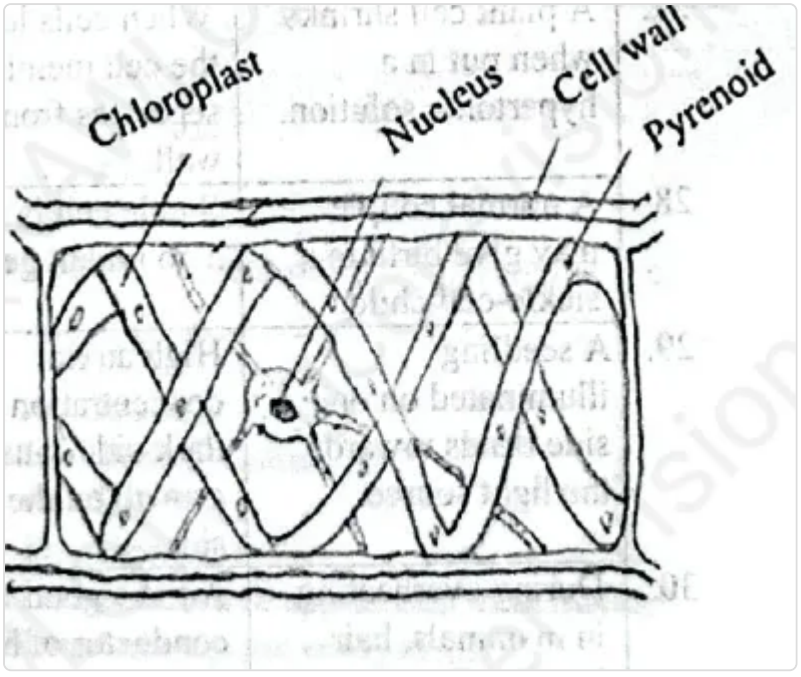
15. Below is the diagram of a bean seed:



During germination, water enters the seed through the:

a) Radicle.
b) Micropyle.
c) Hilum.
d) Testa.

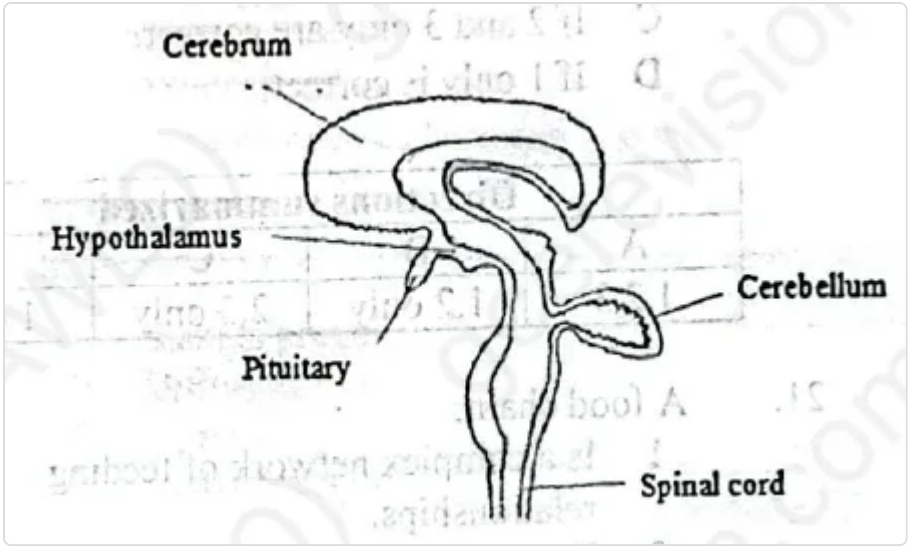
16. The diagram below is that of spirogyra:



Which labelled part enables it to carry out autotrophic nutrition?

a) Chloroplast.
b) Cell wall.
c) Nucleus.
d) Pyrenoid.

17. The diagram below is that of the mammalian brain.



Which of the labelled parts controls reasoning and intelligence?

a) Hypothalamus.
b) Pituitary.
c) Cerebrum.
d) Cerebellum.

18. Which of the following structures in the ileum transports fatty acid and glycerol?

a) Capillaries.
b) Lacteal.
c) Epithelium.
d) Microvilli.

19. Where in the mammalian skin is sebum produced?

a) Malpighian layer.
b) Sebaceous gland.
c) Sweat gland.
d) Granular layer.

20. Carbon dioxide concentration in an industrial zone can be kept constant by:

a) Combustion of fuel.
b) Deforestation.
c) Afforestation.
d) Grazing.

21. *Section III (Questions 21-25). Choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.*

A food chain:

1. Is a complex network of feeding relationships.
2. Has a producer at the base.
3. Is a linear sequence of feeding.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct

c) 2 and 3 only are correct
d) 1 only is correct

22. *Section III (Questions 21-25). Choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.*

The following features maximize absorption of sunlight in leaves:

1. Thin epidermis.
2. Large surface area.
3. Waxy cuticle on lower epidermis.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 1 only is correct

23. *Section III (Questions 21-25). Choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.*

When a complete ring of bark of a tree is peeled off:

1. Active transport of minerals occur.
2. Phloem is destroyed.
3. No translocation of food occurs.

Options as printed: A) 1, 2 and 3 are all correct B) 1 and 2 only are correct C) 2 and 3 only are correct D) 1 only is correct

24. *Section III (Questions 21-25). Choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.*

Decomposers in an ecosystem:

1. Feed saprophytically.
2. Breakdown dead bodies.
3. Release nutrients.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct
c) 2 and 3 only are correct
d) 1 only is correct

25. *Section III (Questions 21-25). Choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.*

In a human being, in the face of danger:

1. Much ADH is secreted.
2. Heart beats faster.
3. Pupil opens wider.

a) 1, 2 and 3 are all correct
b) 1 and 2 only are correct

c) 2 and 3 only are correct
d) 1 only is correct

26. *Section IV (Questions 26-30). Decide whether the first statement is true or false and whether the second statement is true or false. Choose A if both statements are true and the second is a correct explanation of the first; B if both are true and the second is NOT a correct explanation of the first; C if the first is true but the second is false; D if the first is false but the second is true.*

First statement: Oxygen depletion in an aquatic environment results in suffocation and death of aquatic organisms.

Second statement: During eutrophication, aerobic micro-organisms use up large amount of oxygen for decomposition of algae.

a) Both statements are true and the second statement is a correct explanation of the first.
b) Both statements are true and the second statement is NOT a correct explanation of the first.
c) The first statement is true but the second statement is false.
d) The first statement is false but the second statement is true.

27. *Section IV (Questions 26-30). Decide whether the first statement is true or false and whether the second statement is true or false. Choose A if both statements are true and the second is a correct explanation of the first; B if both are true and the second is NOT a correct explanation of the first; C if the first is true but the second is false; D if the first is false but the second is true.*

First statement: A plant cell shrinks when put in a hypertonic solution.

Second statement: When cells lose water, the cell membrane separates from the cell wall.

Options as printed: A) Both statements are true and the second statement is a correct explanation of the first. B) Both statements are true and the second statement is NOT a correct explanation of the first. C) The first statement is true but the second statement is false. D) The first statement is false but the second statement is true.

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First statement: A normal couple may give birth to a sickle-cell child.

Second statement: Sickle cell is caused by a dominant gene.

a) Both statements are true and the second statement is a correct explanation of the first.
b) Both statements are true and the second statement is NOT a correct explanation of the first.
c) The first statement is true but the second statement is false.
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First statement: A seedling illuminated on one side bends towards the light source.

Second statement: High auxin concentration at the dark-side causes rapid growth on the dark-side.

a) Both statements are true and the second statement is a correct explanation of the first.
b) Both statements are true and the second statement is NOT a correct explanation of the first.
c) The first statement is true but the second statement is false.
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First statement: During overheating in mammals, hair lies flat.

Second statement: Air is a good conductor of heat.

a) Both statements are true and the second statement is a correct explanation of the first.
b) Both statements are true and the second statement is NOT a correct explanation of the first.
c) The first statement is true but the second statement is false.
d) The first statement is false but the second statement is true.

31. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Becomes the fruit after fertilization.

a) Stigma.
b) Anther.
c) Ovule.
d) Ovary.

32. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Has both endocrine and exocrine functions.

a) Pituitary gland.
b) Thyroid gland.
c) Adrenal gland.
d) Pancreas.

33. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: An abiotic factor.

a) Pond water.
b) Pond weed.
c) Tadpole.
d) Fish.

34. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: A chemical used to indicate the presence of carbon dioxide.

a) Copper sulphate.
b) Lime water.

c) Sodium bicarbonate.
d) Pyrogallic acid.

35. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Causes an organism to experience orgasm.

a) Ovulation.
b) Copulation.
c) Implantation.
d) Fertilization.

36. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Controls the amount of light entering the eye.

a) Lens.
b) Cornea.
c) Pupil.
d) Iris.

37. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Caused by excessive application of inorganic fertilizers into aquatic ecosystems.

a) Erosion.
b) Desertification.
c) Leaching.
d) Eutrophication.

38. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Is an adaptation for cellulose digestion.

a) Short gut.
b) One-chambered stomach.
c) Sharp chisel-like canines.
d) Presence of bacteria in gut.

39. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: A process which releases Nitrogen into the atmosphere.

a) Nitrogen fixation.
b) Nitrification.
c) Decay.
d) Denitrification.

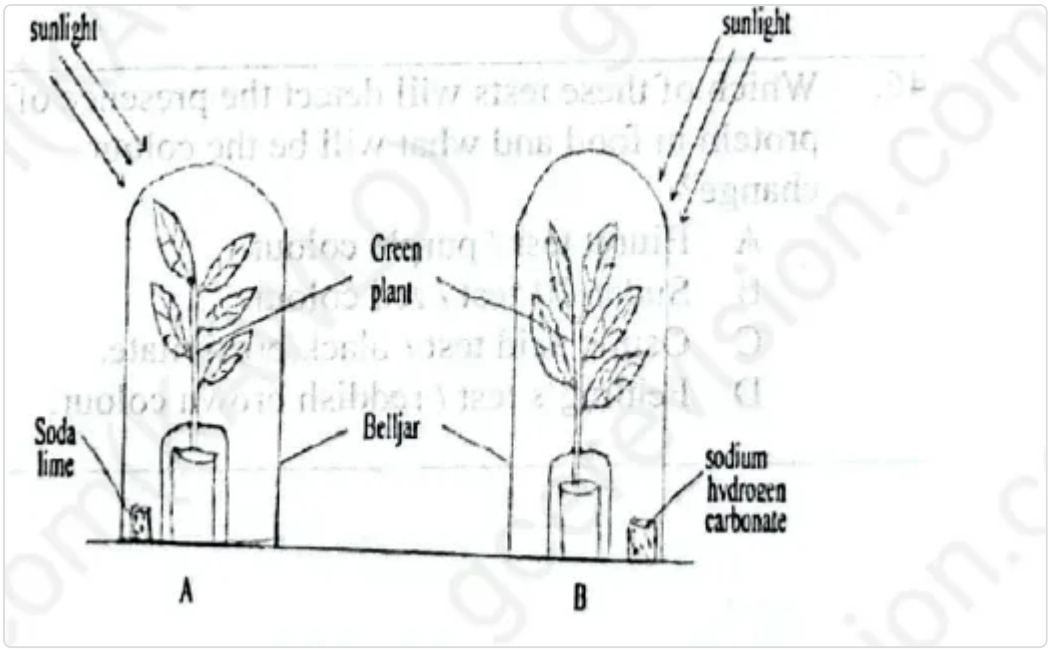
40. *Section V (Questions 31-40). Select the heading which is closely related to the description which follows.*

Description: Movement of substances against their concentration gradient.

a) Suction pressure.
b) Diffusion.

c) Osmosis.
d) Active transport.

41. The apparatus below was set up to investigate the effect of an external factor on photosynthesis.



What is the role of soda lime in the apparatus above?

a) To prove that CO_2 is necessary for photosynthesis.
b) To release CO_2 into the bell jar.
c) To prevent CO_2 production by soil organisms.
d) To absorb CO_2 present in the bell jar.

42. Pyrogallic acid was used in an experiment to show that oxygen is necessary for germination. What is the role of pyrogallic acid in the experiment?

a) To release O_2.
b) To absorb O_2.
c) To kill microorganisms.
d) To facilitate seed germination.

43. Which of the equations below represent the process of fermentation?

Options as printed: A) $6CO_2 + 6H_2O \longrightarrow C_6H_{12}O_6 + 6O_2$. B) $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + Energy. C) $C_6H_{12}O_6 \longrightarrow 2C_3H_5OH + 2CO_2$ + Energy. D) $C_6H_{12}O_6 \longrightarrow 2CH_3CH(OH)COOH$ + Energy.

44. Blood vessels constrict in cold weather such that:

a) Less blood flows to the body surface and conserve heat.
b) More blood flows to the body surface and promote heat loss.
c) More heat is lost by conduction.
d) Less heat is generated by convection.

45. The movement of molecules of a substance from a region of high concentration to a region of low concentration is?

a) Diffusion.
b) Active transport.

c) Osmosis.
d) Mass flow.

46. Which of these tests will detect the presence of protein in food and what will be the colour change?

a) Biuret test / purple colour.
b) Sudan III test / red colour.
c) Osmic acid test / black precipitate.
d) Fehling's test / reddish brown colour.

47. When setting up a potometer, the shoot to be used is cut under water. This is to avoid:

a) Water flowing backward in the xylem.
b) The entry of dust into the xylem.
c) The entry of air into the xylem.
d) The entry of bacteria .

48. Newly transplanted seedlings wilt on a windy sunny day. This is:

a) To reduce excessive water loss.
b) Because they lose water faster than they are absorbing.
c) Because the guard cells become very turgid.
d) Because the seedlings do not yet have a thick cuticle to resist transpiration.

49. In an experiment to determine a change in mass over time, a fixed quantity of germinating seeds were placed in a beaker and weighed on a balance after time intervals. After a few days, a drop in mass was noticed. The drop in mass at the early days was due to:

a) No respiration.
b) Digestion and use of food reserves.
c) Inability to absorb water.
d) Digestion of the testa.

50. Brown eye (B) is dominant over blue eye (b). A heterozygous brown-eyed man marries a blue-eyed woman. What is the probability of this couple producing blue-eyed offspring?

a) $\frac{1}{2}$.
b) $\frac{3}{4}$.
c) 0.
d) $\frac{1}{4}$.

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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